

Project Summary: StockMarket App Deployment (Ansible, Terraform, Cloud Architecture)

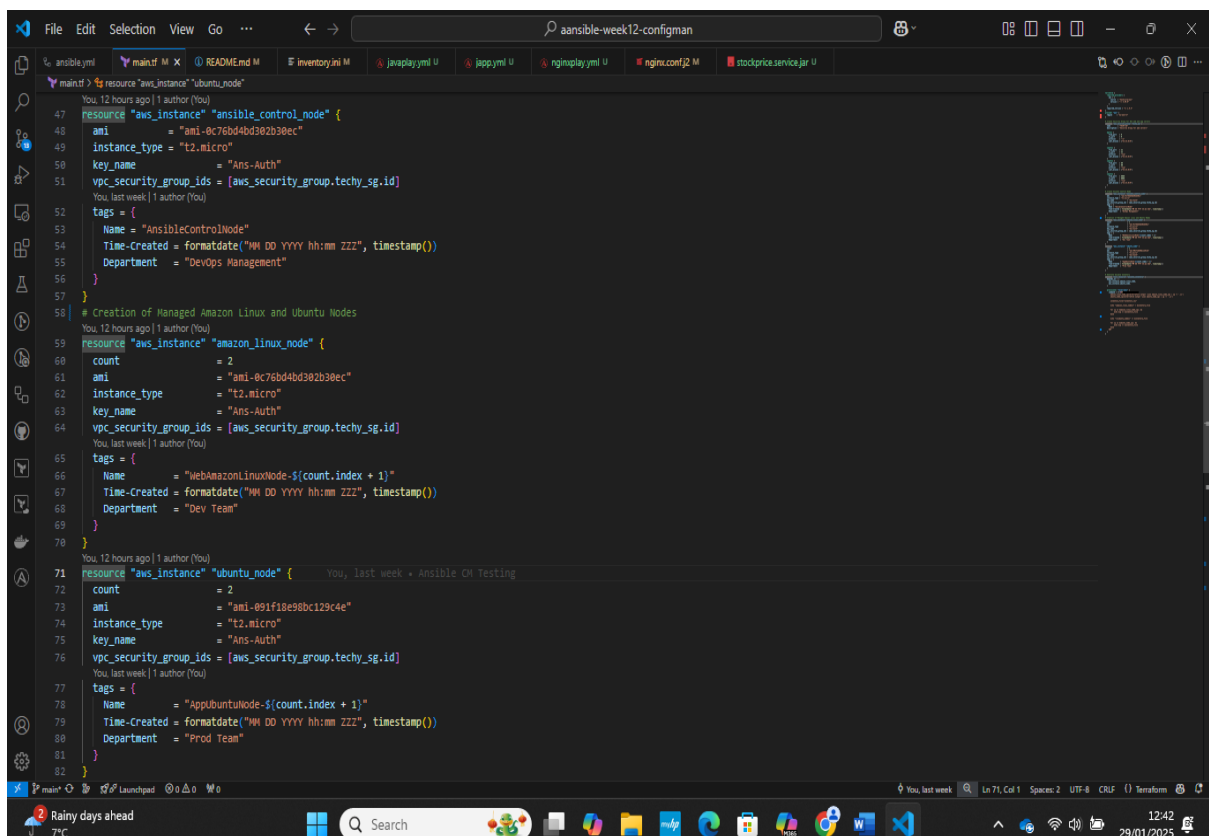
As the **DevOps Engineer** for this project, I successfully deployed the **StockMarket App**, which comprises **two Java backend systems** and **two Nginx reverse proxy servers**, all hosted within a **cloud infrastructure**. The project was structured into **three distinct levels**, focusing on **infrastructure provisioning**, **automation of configurations**, and ensuring **efficient traffic routing** between Nginx and Java services.

Step-by-Step Implementation

Level 1: Infrastructure Setup (Automated with Terraform)

✅ **Provision 5 Nodes (2 Nginx, 2 Java, 1 Database/Worker Node)** You can use **Terraform** to automate cloud instance creation or manually deploy them.

Terraform-Based Node Creation (main.tf)

A screenshot of a code editor window displaying a Terraform configuration file named 'main.tf'. The editor has a dark theme and shows several tabs at the top: 'ansible.yml', 'main.tf', 'README.md', 'inventory.ini', 'javaplay.yml', 'japp.yml', 'nginxplay.yml', 'nginx.conf', 'stockprice.service.jar'. The 'main.tf' tab is active, showing Terraform code for creating AWS instances. The code includes comments and resource definitions for 'aws_instance' with various attributes like 'ami', 'instance_type', 'key_name', 'vpc_security_group_ids', and 'tags'. The configuration is organized into three main sections: creating an Ansible control node, creating two Amazon Linux nodes, and creating two Ubuntu nodes. The bottom of the image shows a Windows taskbar with various application icons and a system clock indicating 12:42 on 29/01/2025.

```
47 resource "aws_instance" "ansible_control_node" {
48   ami           = "ami-0c76bd4bd302b30ec"
49   instance_type = "t2.micro"
50   key_name      = "Ans-Auth"
51   vpc_security_group_ids = [aws_security_group.techy_sg.id]
52   tags = {
53     Name        = "AnsibleControlNode"
54     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
55     Department  = "DevOps Management"
56   }
57 }
58 # Creation of Managed Amazon Linux and Ubuntu Nodes
59 resource "aws_instance" "amazon_linux_node" {
60   count          = 2
61   ami           = "ami-0c76bd4bd302b30ec"
62   instance_type = "t2.micro"
63   key_name      = "Ans-Auth"
64   vpc_security_group_ids = [aws_security_group.techy_sg.id]
65   tags = {
66     Name        = "WebAmazonLinuxNode-${count.index + 1}"
67     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
68     Department  = "Dev Team"
69   }
70 }
71 resource "aws_instance" "ubuntu_node" {
72   count          = 2
73   ami           = "ami-091f18e98bc129c4e"
74   instance_type = "t2.micro"
75   key_name      = "Ans-Auth"
76   vpc_security_group_ids = [aws_security_group.techy_sg.id]
77   tags = {
78     Name        = "AppUbuntuNode-${count.index + 1}"
79     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
80     Department  = "Prod Team"
81   }
82 }
```

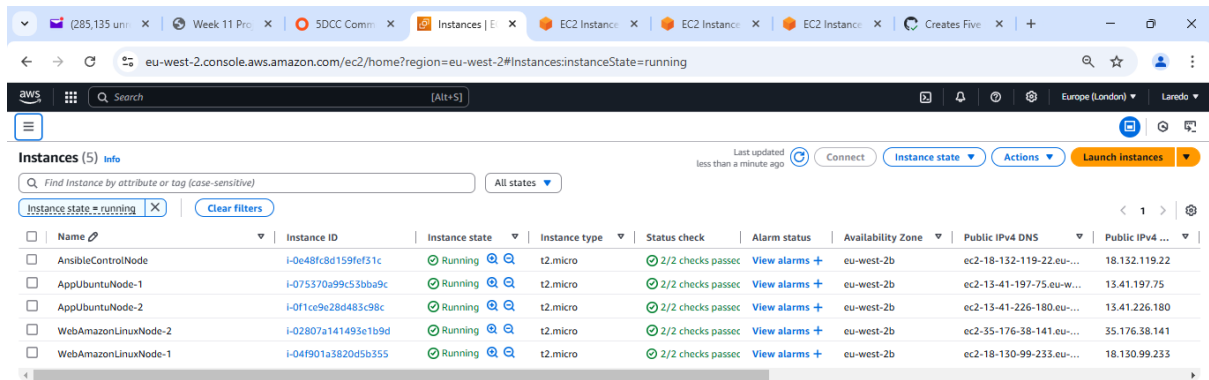
GitHub Actions Workflow

```
File Edit Selection View Go ... ansible-week12-configman
ansible.yml
1 You, 16 seconds ago | 1 author (You)
2 name: Deploy AWS EC2 Instances
3
4 on:
5   push:
6     branches:
7       - main
8   workflow_dispatch:
9     inputs:
10      action:
11        description: 'This enforces Separation of Concerns'
12        required: true
13        default: 'apply'
14        type: choice
15        options:
16          - apply
17          - destroy
18
19 env:
20   AWS_ACCESS_KEY_ID: ${ secrets.AWS_ACCESS_KEY_ID }
21   AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_KEY_ID }
22   AWS_DEFAULT_REGION: eu-west-2
23
24 jobs:
25   deploy:
26     runs-on: ubuntu-latest
27
28     steps:
29       - name: Checkout code
30         uses: actions/checkout@v2
31
32       - name: Setup Terraform
33         uses: hashicorp/setup-terraform@v1
34
35       - name: Terraform Init
36         run: terraform init
37
38       - name: Terraform Validate
39         run: terraform validate
40
41       - name: Terraform Plan
42         id: plan
43         run: terraform plan -out-plan.tfplan
44
45       - name: Convert Terraform Output to JSON
46         run: |
47           terraform output -json > inventory.json
48           cat inventory.json | tee inventory.ini # Output the content for debugging
49
50       - name: Terraform Apply
51         if: github.event.name == 'push' || (github.event.inputs.action == 'apply')
52         run: terraform apply -auto-approve plan.tfplan
53
54       - name: Terraform Destroy
55         if: github.event.name == 'push' || (github.event.inputs.action == 'destroy')
56         run: terraform destroy -auto-approve
```

Workflow success

```
File Edit Selection View Go ... ansible-week12-configman
main.tf
47 resource "aws_instance" "ansible_control_node" {
48   ami           = "ami-0c76bd4bd302b30ec"
49   instance_type = "t2.micro"
50   key_name      = "Ans-Auth"
51   vpc_security_group_ids = [aws_security_group.techy_sg.id]
52   tags = {
53     Name = "AnsibleControlNode"
54     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
55     Department = "DevOps Management"
56   }
57 }
58 # Creation of Managed Amazon Linux and Ubuntu Nodes
59 resource "aws_instance" "amazon_linux_node" {
60   count          = 2
61   ami           = "ami-0c76bd4bd302b30ec"
62   instance_type = "t2.micro"
63   key_name      = "Ans-Auth"
64   vpc_security_group_ids = [aws_security_group.techy_sg.id]
65   tags = {
66     Name = "WebAmazonLinuxNode-${count.index + 1}"
67     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
68     Department = "Dev Team"
69   }
70 }
71 resource "aws_instance" "ubuntu_node" {
72   count          = 2
73   ami           = "ami-091f18e98bc129c4e"
74   instance_type = "t2.micro"
75   key_name      = "Ans-Auth"
76   vpc_security_group_ids = [aws_security_group.techy_sg.id]
77   tags = {
78     Name = "AppUbuntuNode-${count.index + 1}"
79     Time-Created = formatdate("MM DD YYYY hh:mm ZZ", timestamp())
80     Department = "Prod Team"
81   }
82 }
```

AWS Console showing Nodes Provisioned via Terraform



Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
AnsibleControlNode	i-0e48f8d159fef31c	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-2b	ec2-18-132-119-22.eu-...	18.132.119.22
AppUbuntuNode-1	i-075370a99c53bba9c	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-2b	ec2-13-41-197-75.eu-...	13.41.197.75
AppUbuntuNode-2	i-0f1ce9e28d483c98c	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-2b	ec2-13-41-226-180.eu-...	13.41.226.180
WebAmazonLinuxNode-2	i-02807a141493e1b9d	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-2b	ec2-35-176-38-141.eu-...	35.176.38.141
WebAmazonLinuxNode-1	i-04f901a3820d5b355	Running	t2.micro	2/2 checks passed	View alarms +	eu-west-2b	ec2-18-130-99-233.eu-...	18.130.99.233

Select an instance



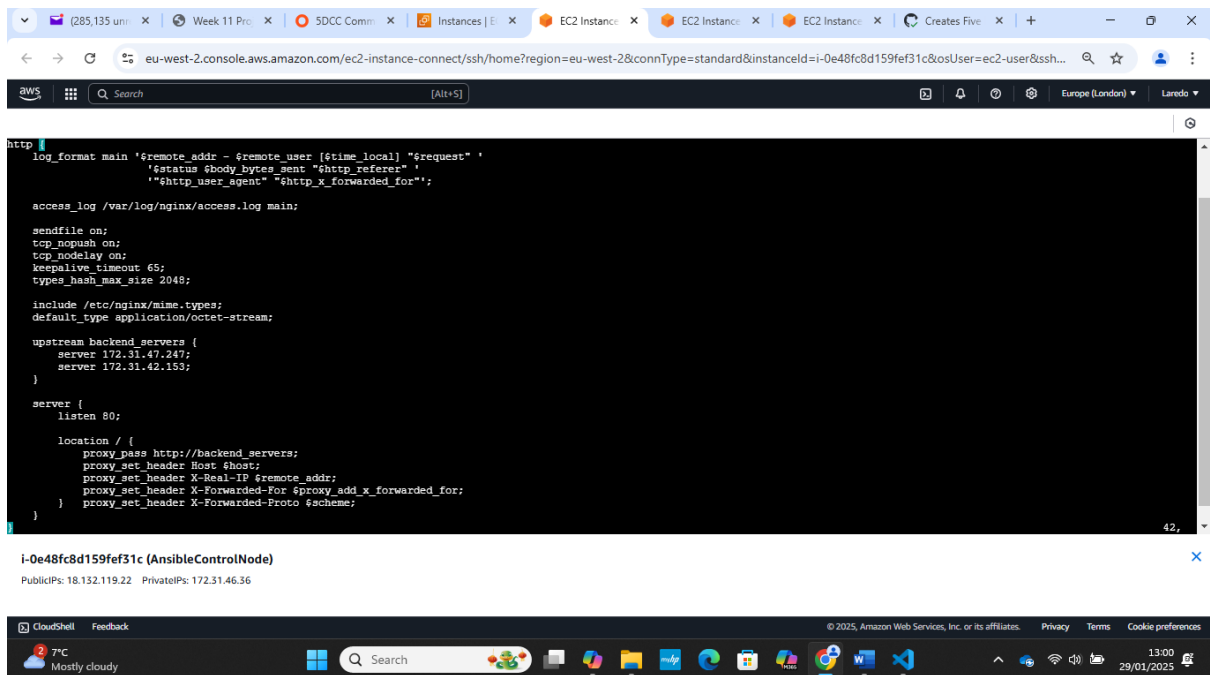
Level 2: Nginx Configuration with Ansible

✓ Install and Configure Nginx

Playbook to install Nginx (and Configure as REVERSE PROXY) on Nodes

```
File Edit Selection View Go ... ansible-week12-configman
1 - name: Install and configure Nginx as a reverse proxy
2   hosts: web
3   become: yes
4   tasks:
5     - name: Update all packages on Amazon Linux
6       yum:
7         name: '*'
8         state: latest
9       when: ansible_facts['distribution'] in ['Amazon', 'Redhat']
10
11    - name: Update all packages on Ubuntu and Debian
12      apt:
13        update_cache: yes
14        upgrade: dist
15      when: ansible_facts['distribution'] in ['Ubuntu', 'Debian']
16
17    - name: Install Nginx on Amazon Linux
18      yum:
19        name: nginx
20        state: present
21      when: ansible_facts['distribution'] in ['Amazon', 'Redhat']
22
23    - name: Install Nginx on Ubuntu
24      apt:
25        name: nginx
26        state: present
27        update_cache: yes
28      when: ansible_facts['distribution'] in ['Ubuntu', 'Debian']
29
30    - name: Start and enable Nginx service
31      service:
32        name: nginx
33        state: started
34        enabled: yes
35
36    - name: Configure Nginx as a reverse proxy
37      template:
38        src: nginx.conf.j2
39        dest: /etc/nginx/nginx.conf
40      notify:
41        - reload nginx
42
43    handlers:
44      - name: reload nginx
45        service:
46          name: nginx
47          state: reloaded
```

✓ Nginx Reverse Proxy Template (nginx.conf.j2)



The screenshot shows a terminal window within the AWS Management Console. The terminal displays the content of the Nginx configuration file, which is a template for a reverse proxy. The configuration includes settings for log format, access log, sendfile, tcp_nopush, tcp_nodelay, keepalive_timeout, types_hash_max_size, include /etc/nginx/mime.types, default_type application/octet-stream, upstream backend_servers, and a server block listening on port 80. The server block has a location / that proxies requests to the backend servers, setting various headers like Host, X-Real-IP, X-Forwarded-For, and X-Forwarded-Proto.

```
http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    tcp_nodelay on;
    keepalive_timeout 65;
    types_hash_max_size 2048;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    upstream backend_servers {
        server 172.31.47.247;
        server 172.31.42.153;
    }

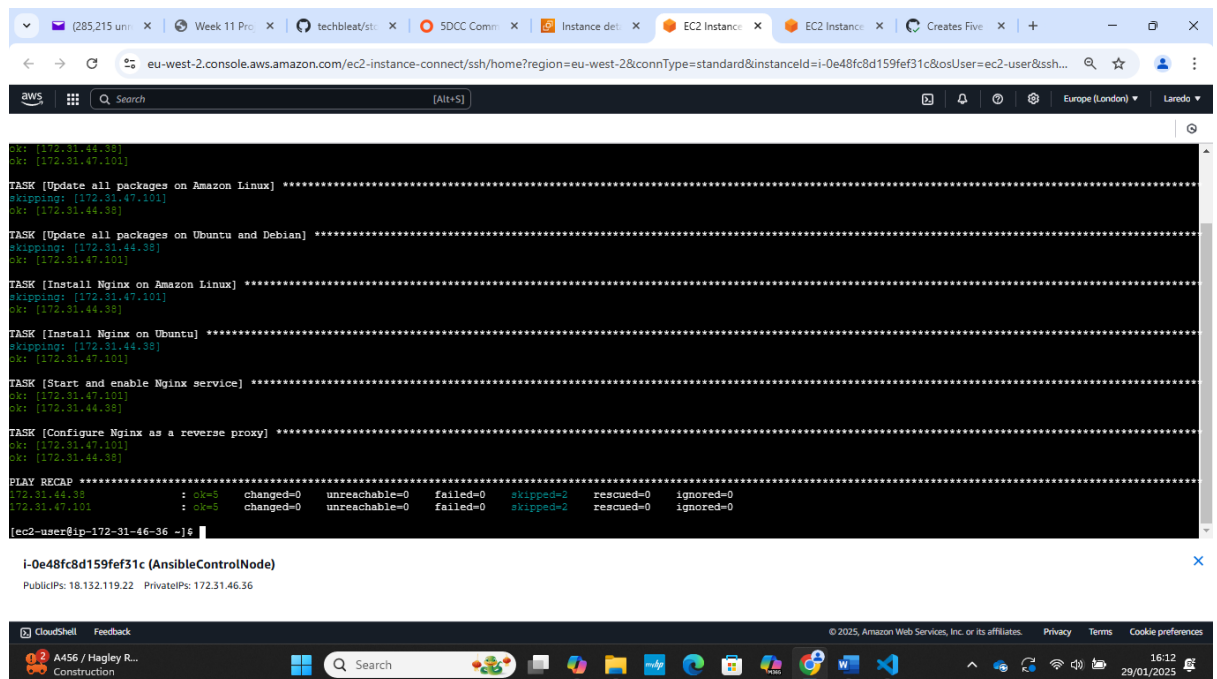
    server {
        listen 80;

        location / {
            proxy_pass http://backend_servers;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
        }
    }
}
```

i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 18.132.119.22 PrivateIPs: 172.31.46.36

✓ Run: ansible-playbook

Nginx service setup as a REVERSE PROXY for the Java service: Success



The screenshot shows a terminal window within the AWS Management Console displaying the output of an Ansible playbook run. The output shows the execution of several tasks, including updating packages, installing Nginx, and configuring it as a reverse proxy. The tasks are executed on the AnsibleControlNode. The output also shows a summary of the results, indicating that all tasks were successful.

```
ok: [172.31.44.38]
ok: [172.31.47.101]

TASK [Update all packages on Amazon Linux] *****
skipping: [172.31.47.101]
ok: [172.31.44.38]

TASK [Update all packages on Ubuntu and Debian] *****
skipping: [172.31.44.38]
ok: [172.31.47.101]

TASK [Install Nginx on Amazon Linux] *****
skipping: [172.31.47.101]
ok: [172.31.44.38]

TASK [Install Nginx on Ubuntu] *****
skipping: [172.31.44.38]
ok: [172.31.47.101]

TASK [Start and enable Nginx service] *****
ok: [172.31.47.101]
ok: [172.31.44.38]

TASK [Configure Nginx as a reverse proxy] *****
ok: [172.31.47.101]
ok: [172.31.44.38]

PLAY RECAP *****
172.31.44.38 : ok=5 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0
172.31.47.101 : ok=5 changed=0 unreachable=0 failed=0 skipped=2 rescued=0 ignored=0

[ec2-user@ip-172-31-46-36 ~]$
```

i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 18.132.119.22 PrivateIPs: 172.31.46.36

Reverse proxy for WebAmazonLinuxNode-1 (also AppUbuntuNode-1)

285,215 unread

Week 11 Project

Instance details

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eu-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-2&connType=standard&instanceId=i-04f901a3820d5b355&osUser=ec2-user&ss...

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Europe (London)

Laredo

```
"$http_user_agent" "$http_x_forwarded_for";

access_log /var/log/nginx/access.log main;

sendfile on;
tcp_nopush on;
tcp_nodelay on;
keepalive_timeout 65;
types_hash_max_size 2048;

include /etc/nginx/mime.types;
default_type application/octet-stream;

upstream backend_servers {
    server 172.31.47.247:8080;
    server 172.31.42.153:8080;
}

server {
    listen 80;

    location / {
        proxy_pass http://backend_servers;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

i-04f901a3820d5b355 (WebAmazonLinuxNode-1)

PublicIPs: 18.130.99.233 PrivateIPs: 172.31.44.38

CloudShell Feedback

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77°C Cloudy

Search

16:16 29/01/2025

Nginx service running on WebAmazonLinuxNode-1 (also AppUbuntuNode-1)

285,215 unread

Week 11 Project

Instance details

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eu-west-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-west-2&connType=standard&instanceId=i-04f901a3820d5b355&osUser=ec2-user&ss...

Q Search [Alt+S]

Europe (London)

Laredo

```
[ec2-user@ip-172-31-44-38 ~]$ nginx -version
nginx version: nginx/1.26.2
[ec2-user@ip-172-31-44-38 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-01-29 00:57:52 UTC; 15h ago
     Process: 7640 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 7674 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 7710 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
     Process: 58778 ExecReload=/usr/sbin/nginx -s reload (code=exited, status=0/SUCCESS)
    Main PID: 7765 (nginx)
      Tasks: 2 (limit: 1111)
     Memory: 2.7M
           CPU: 145ms
    CGroup: /system.slice/nginx.service
            └─ 7765 "nginx: master process /usr/sbin/nginx"
               └─ 58779 "nginx: worker process"

Jan 29 00:57:52 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Jan 29 00:57:52 ip-172-31-44-38.eu-west-2.compute.internal nginx[7674]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Jan 29 00:57:52 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
Jan 29 00:57:55 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Reloading nginx.service - The nginx HTTP and reverse proxy server...
Jan 29 00:57:55 ip-172-31-44-38.eu-west-2.compute.internal nginx[11392]: nginx: [warn] could not build optimal types_hash, you should increase either types_hash_max_size: 2048 or types_
Jan 29 00:57:55 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Reloaded nginx.service - The nginx HTTP and reverse proxy server.
Jan 29 13:41:32 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Reloading nginx.service - The nginx HTTP and reverse proxy server...
Jan 29 13:41:32 ip-172-31-44-38.eu-west-2.compute.internal nginx[58778]: nginx: [warn] could not build optimal types_hash, you should increase either types_hash_max_size: 2048 or types_
Jan 29 13:41:32 ip-172-31-44-38.eu-west-2.compute.internal systemd[1]: Reloaded nginx.service - The nginx HTTP and reverse proxy server.
lines 1-25/25 (END)
```

i-04f901a3820d5b355 (WebAmazonLinuxNode-1)

PublicIPs: 18.130.99.233 PrivateIPs: 172.31.44.38

CloudShell Feedback

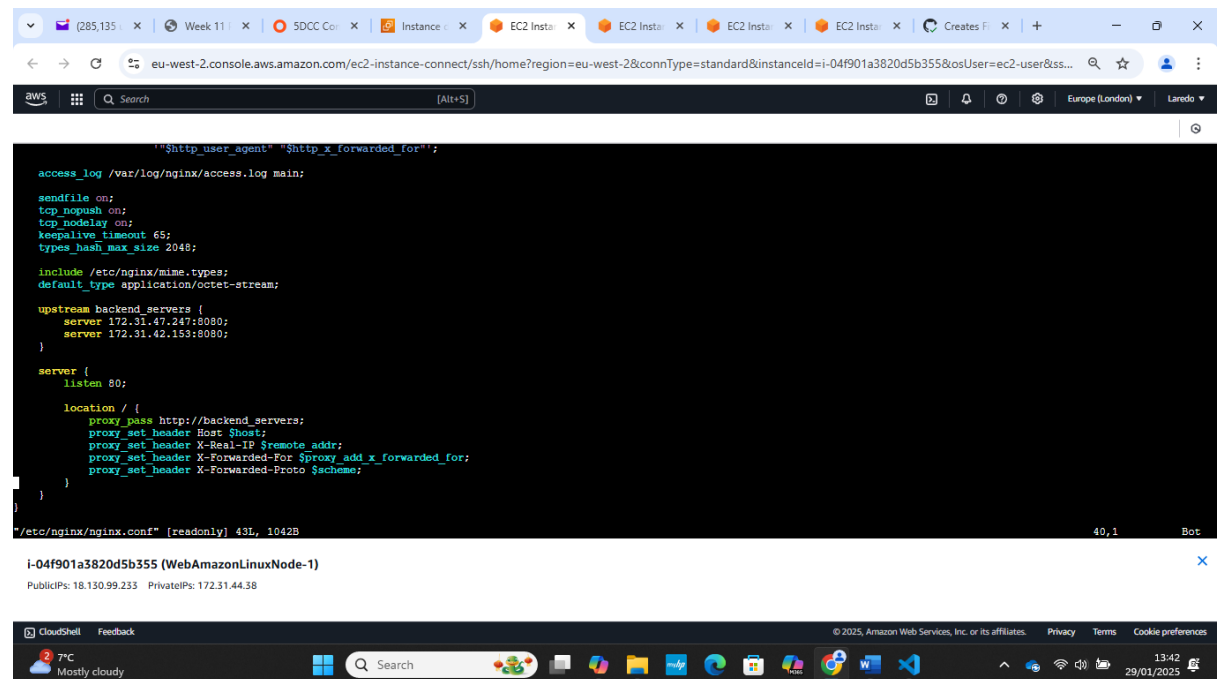
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77°C Cloudy

Search

16:18 29/01/2025

Nginx Reverse proxies forward traffic to the two Java backend nodes on port 8080



The screenshot shows an AWS CloudShell terminal window. The top bar displays the AWS logo, a search bar, and the region 'Europe (London)'. The terminal content shows the Nginx configuration file `/etc/nginx/nginx.conf` with the following settings:

```
"$http_user_agent" "$http_x_forwarded_for";

access_log /var/log/nginx/access.log main;

sendfile on;
tcp_nopush on;
tcp_nodelay on;
keepalive_timeout 65;
types_hash_max_size 2048;

include /etc/nginx/mime.types;
default_type application/octet-stream;

upstream backend_servers {
    server 172.31.47.247:8080;
    server 172.31.42.153:8080;
}

server {
    listen 80;

    location / {
        proxy_pass http://backend_servers;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

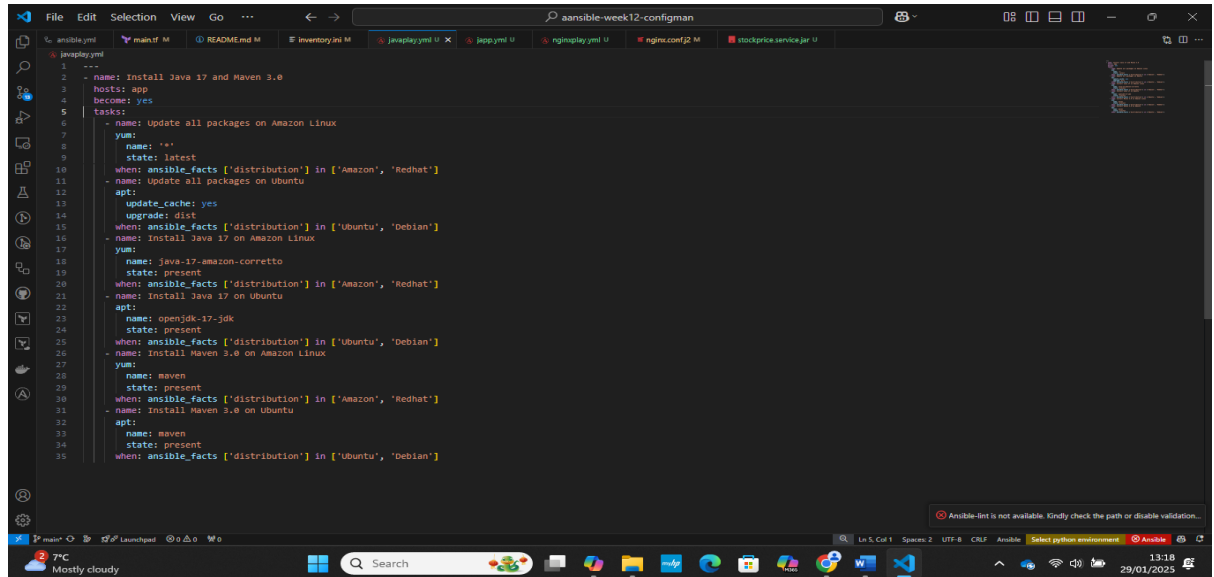
Below the configuration, the instance details for `i-04f901a3820d5b355 (WebAmazonLinuxNode-1)` are shown, including Public IPs (18.130.99.233, 172.31.44.38) and Private IPs (172.31.44.38).

The bottom bar of the CloudShell shows the temperature (7°C, Mostly cloudy), a search bar, and the date/time (13:42, 29/01/2025).

Level 3: Java Backend Deployment with Ansible

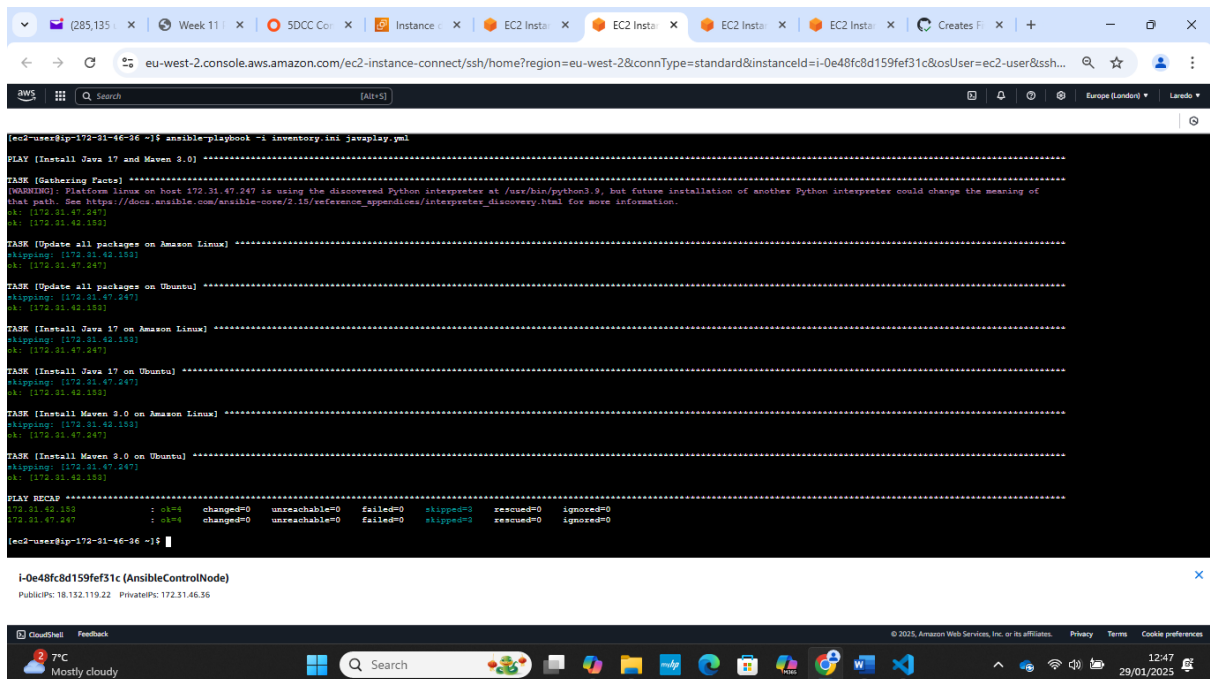
✅ Install Java & Maven

Playbook to install and configure Java 17 and Maven 3.0 on Backend Java nodes.



```
1 ---
2 - name: Install Java 17 and Maven 3.0
3   hosts: app
4   become: yes
5   tasks:
6     - name: Update all packages on Amazon Linux
7       yum:
8         name: '*'
9         state: latest
10      when: ansible_facts['distribution'] in ['Amazon', 'Redhat']
11     - name: Update all packages on Ubuntu
12       apt:
13         update_cache: yes
14         upgrade: dist
15      when: ansible_facts['distribution'] in ['Ubuntu', 'Debian']
16     - name: Install Java 17 on Amazon Linux
17       yum:
18         name: java-17-amazon-corretto
19         state: present
20      when: ansible_facts['distribution'] in ['Amazon', 'Redhat']
21     - name: Install Java 17 on Ubuntu
22       apt:
23         name: openjdk-17-jdk
24         state: present
25      when: ansible_facts['distribution'] in ['Ubuntu', 'Debian']
26     - name: Install Maven 3.0 on Amazon Linux
27       yum:
28         name: maven
29         state: present
30      when: ansible_facts['distribution'] in ['Amazon', 'Redhat']
31     - name: Install Maven 3.0 on Ubuntu
32       apt:
33         name: maven
34         state: present
35      when: ansible_facts['distribution'] in ['Ubuntu', 'Debian']
```

Java & Maven Installed on Amazon and Ubuntu backend App nodes.



```
[ec2-user@ip-172-31-46-36 ~]$ ansible-playbook -i inventory.ini java.yml

PLAY [Install Java 17 and Maven 3.0] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.47.247 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.12/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.47.247]

TASK [Update all packages on Amazon Linux] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]

TASK [Update all packages on Ubuntu] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

TASK [Install Java 17 on Amazon Linux] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]

TASK [Install Java 17 on Ubuntu] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

TASK [Install Maven 3.0 on Amazon Linux] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]

TASK [Install Maven 3.0 on Ubuntu] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

PLAY RECAP *****
ok=6 changed=0 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
[ec2-user@ip-172-31-46-36 ~]$
```

i-0e48fc8d159fef31c (AnsibleControlNode)
Public IP: 18.132.119.22 Private IP: 172.31.46.36

✓ Deploy Java Application

Playbook to deploy the Java app on the Java Backend nodes

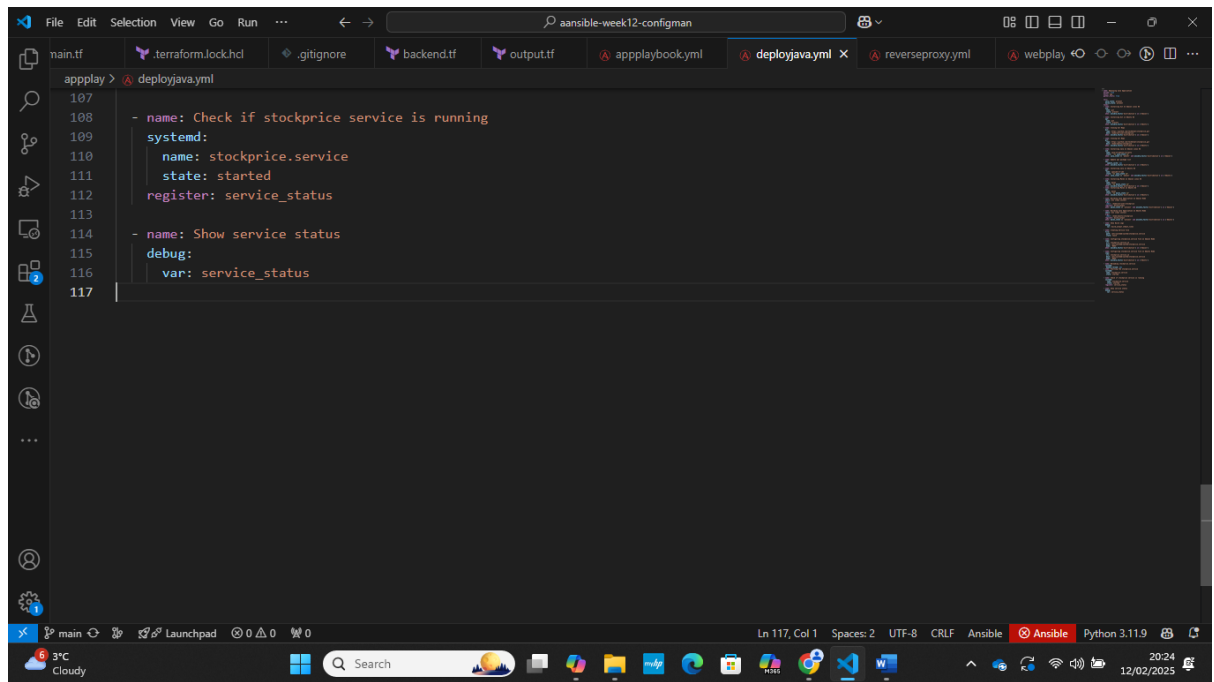
```
1 ---
2 name: Deploying Java Application
3 become: true
4 hosts: app
5 gather_facts: true
6
7 vars:
8   java_state: present
9   maven_state: present
10 tasks:
11   - name: Installing Git In Amazon Linux OS
12     yum:
13       name: git
14       state: present
15     when: ansible_facts['distribution'] in ['Amazon']
16
17   - name: Installing Git in Ubuntu OS
18     apt:
19       name: git
20       state: present
21     when: ansible_facts['distribution'] in ['Ubuntu']
22
23   - name: Cloning Git Repo
24     git:
25       repo: https://github.com/techbleat/stockprice.git
26       dest: /home/ec2-user/stockprice
27     when: ansible_facts['distribution'] in ['Amazon']
```

```
29   - name: Cloning Git Repo
30     git:
31       repo: https://github.com/techbleat/stockprice.git
32       dest: /home/ubuntu/stockprice
33     when: ansible_facts['distribution'] in ['Ubuntu']
34
35   - name: Installing Java on Amazon Linux OS
36     yum:
37       name: java-17-amazon-corretto
38       state: "{{ java_state }}"
39     when: java_state == 'absent' and ansible_facts['distribution'] in ['Amazon']
40
41   - name: Update apt package list
42     apt:
43       update_cache: yes
44     when: ansible_facts['distribution'] in ['Ubuntu']
45
46   - name: Installing Java on Ubuntu OS
47     apt:
48       name: openjdk-17-jdk
49       state: "{{ java_state }}"
50     when: java_state == 'absent' and ansible_facts['distribution'] in ['Ubuntu']
51
52   - name: Installing Maven on Amazon Linux OS
53     yum:
54       name: maven
55       state: "{{ maven_state }}"
56     when: ansible_facts['distribution'] in ['Amazon']
```

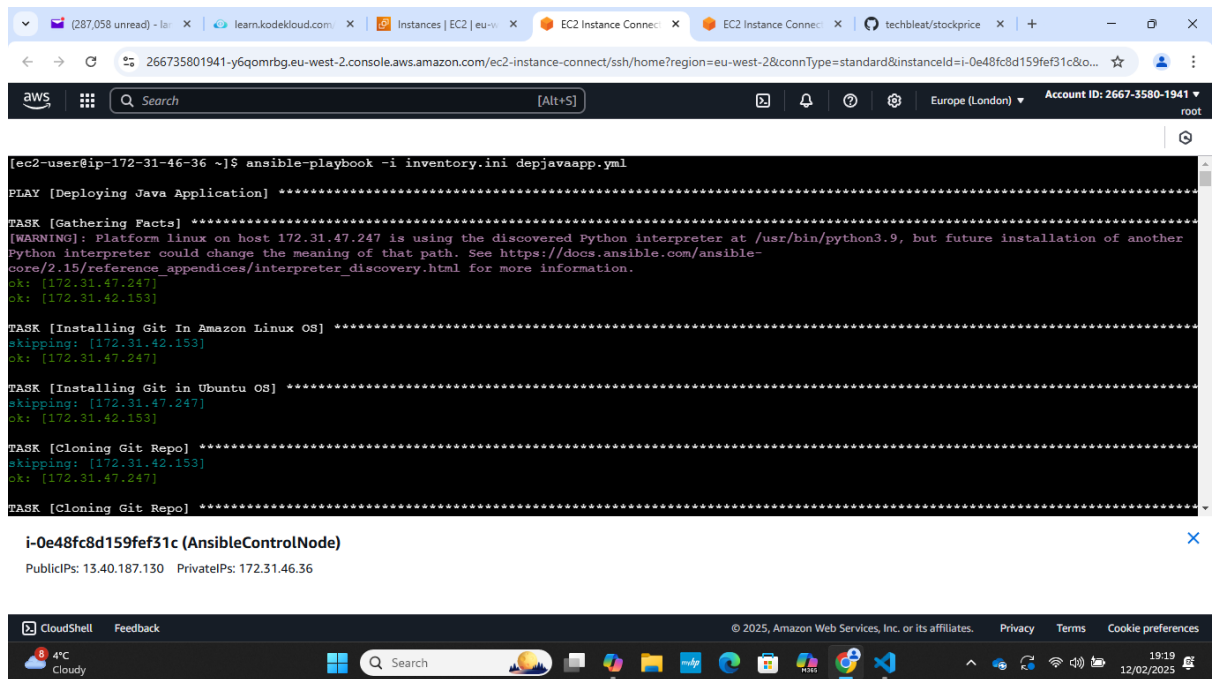


```
62
63 - name: Building Java Application on Amazon Node
64   shell: mvn clean install
65   args:
66     chdir: /home/ec2-user/stockprice
67     register: build_output
68     when: maven_state == 'present' and ansible_facts['distribution'] in ['Amazon']
69
70 - name: Building Java Application on Ubuntu Node
71   shell: mvn clean install
72   args:
73     chdir: /home/ubuntu/stockprice
74     register: build_output
75     when: maven_state == 'present' and ansible_facts['distribution'] in ['Ubuntu']
76
77 - name: Show Build Logs
78   debug:
79     var: build_output.stdout_lines
80
81 - name: Creating Service file
82   file:
83     path: /etc/systemd/system/stockprice.service
84     state: touch
85
86 - name: Configuring stockprice.service file on Amazon Node
87   copy:
88     src: stockprice.service.j2
89     dest: /etc/systemd/system/stockprice.service
90     mode: '0644'
91     when: ansible_facts['distribution'] in ['Amazon']
92
93 - name: Configuring stockprice.service file on Ubuntu Node
94   copy:
95     src: stockprice.service.j2
96     dest: /etc/systemd/system/stockprice.service
97     mode: '0644'
98     when: ansible_facts['distribution'] in ['Ubuntu']
99
100 - name: Reloading stockprice.service
101   systemd:
102     daemon_reload: yes
103
104 - name: Starting the stockprice.service
105   systemd:
106     name: stockprice.service
107     state: started
108
109 - name: Check if stockprice service is running
110   systemd:
111     name: stockprice.service
112     state: started
113   register: service_status
```

```
85
86 - name: Configuring stockprice.service file on Amazon Node
87   copy:
88     src: stockprice.service.j2
89     dest: /etc/systemd/system/stockprice.service
90     mode: '0644'
91     when: ansible_facts['distribution'] in ['Amazon']
92
93 - name: Configuring stockprice.service file on Ubuntu Node
94   copy:
95     src: stockprice.service.j2
96     dest: /etc/systemd/system/stockprice.service
97     mode: '0644'
98     when: ansible_facts['distribution'] in ['Ubuntu']
99
100 - name: Reloading stockprice.service
101   systemd:
102     daemon_reload: yes
103
104 - name: Starting the stockprice.service
105   systemd:
106     name: stockprice.service
107     state: started
108
109 - name: Check if stockprice service is running
110   systemd:
111     name: stockprice.service
112     state: started
113   register: service_status
```



✓ Run: ansible-playbook for Deployment



```
TASK [Cloning Git Repo] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]

TASK [Cloning Git Repo] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

TASK [Installing Java on Amazon Linux OS] *****
skipping: [172.31.47.247]
skipping: [172.31.42.153]

TASK [Update apt package list] *****
skipping: [172.31.47.247]
changed: [172.31.42.153]

TASK [Installing Java on Ubuntu OS] *****
skipping: [172.31.47.247]
skipping: [172.31.42.153]

TASK [Installing Maven on Amazon Linux OS] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]
```

i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36

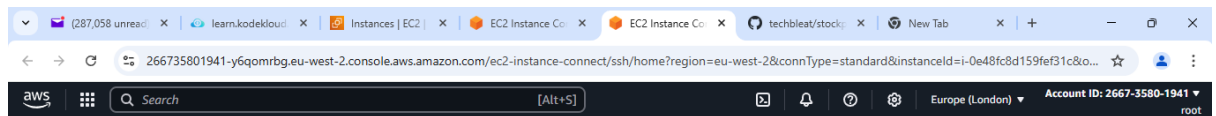
```
TASK [Installing Maven on Ubuntu OS] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

TASK [Building Java Application] *****
skipping: [172.31.42.153]
changed: [172.31.47.247]

TASK [Building Java Application] *****
skipping: [172.31.47.247]
changed: [172.31.42.153]

TASK [Show Build Logs] *****
ok: [172.31.47.247] => {
  "build_output.stdout_lines": "VARIABLE IS NOT DEFINED!"
}
ok: [172.31.42.153] => {
  "build_output.stdout_lines": [
    "[\u001b[1;34mINFO\u001b[m] Scanning for projects...",
    "[\u001b[1;34mINFO\u001b[m] ",
    "[\u001b[1;34mINFO\u001b[m] \u001b[1m-----< \u001b[0;36mcom.techbleat:stock-service\u001b[0;1m >-----\u001b[m]",
    "[\u001b[1;34mINFO\u001b[m] \u001b[1mBuilding stock-service 0.0.1-SNAPSHOT\u001b[m]",
    "[\u001b[1;34mINFO\u001b[m] \u001b[1m-----[ jar ]-----\u001b[m]",
  ]
}
```

i-0e48fc8d159fef31c (AnsibleControlNode)
PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36



```
TASK [Creating Service file] *****
changed: [172.31.42.153]
changed: [172.31.47.247]

TASK [Configuring stockprice.service file on Amazon Node] *****
skipping: [172.31.42.153]
ok: [172.31.47.247]

TASK [Configuring stockprice.service file on Ubuntu Node] *****
skipping: [172.31.47.247]
ok: [172.31.42.153]

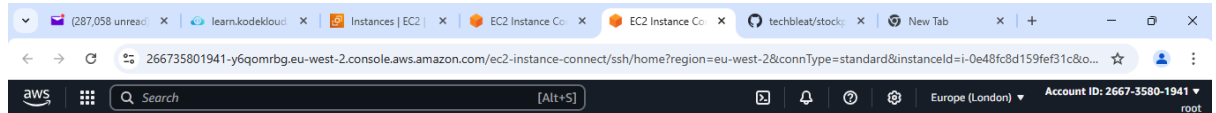
TASK [Reloading stockprice.service] *****
ok: [172.31.42.153]
ok: [172.31.47.247]

TASK [Starting the stockprice.service] *****
changed: [172.31.42.153]
changed: [172.31.47.247]

TASK [Check if stockprice service is running] *****
ok: [172.31.42.153]
ok: [172.31.47.247]
```

i-0e48fc8d159fef31c (AnsibleControlNode)

PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36



```
changed: [172.31.47.247]

TASK [Check if stockprice service is running] *****
ok: [172.31.42.153]
ok: [172.31.47.247]

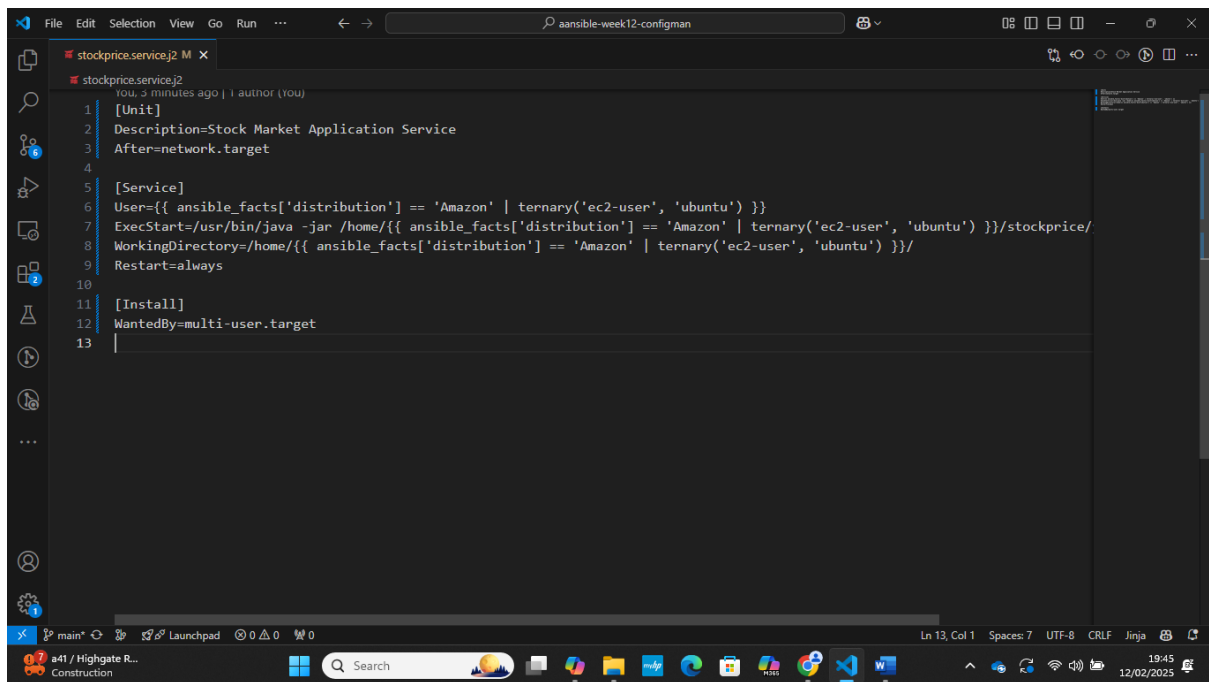
TASK [Show service status] *****
ok: [172.31.47.247] => {
  "service_status": {
    "changed": false,
    "failed": false,
    "name": "stockprice.service",
    "state": "started",
    "status": {
      "AccessSELinuxContext": "system_u:object_r:systemd_unit_file_t:s0",
      "ActiveEnterTimestamp": "Wed 2025-02-12 20:16:09 UTC",
      "ActiveEnterTimestampMonotonic": "22434330709",
      "ActiveExitTimestamp": "Wed 2025-02-12 20:16:09 UTC",
      "ActiveExitTimestampMonotonic": "22434331178",
      "ActiveState": "activating",
      "After": "network.target systemd-journald.socket basic.target sysinit.target system.slice -.mount",
      "AllowIsolate": "no",
      "AssertResult": "yes",
      "AssertTimestamp": "Wed 2025-02-12 20:16:08 UTC",
```

i-0e48fc8d159fef31c (AnsibleControlNode)

PublicIPs: 13.40.187.130 PrivateIPs: 172.31.46.36



Created a service wrapper around the java application to easily manage it

A screenshot of a code editor window showing an Ansible playbook named 'stockprice.servicej2'. The playbook is written in YAML and includes sections for [Unit], [Service], and [Install]. The [Service] section defines the user, exec start command, working directory, and restart policy. The [Install] section defines the wanted by target. The editor has a dark theme and a sidebar on the left with icons for file explorer, search, and other tools. The bottom status bar shows the current line and column, encoding, and other details.

```
1 [Unit]
2 Description=Stock Market Application Service
3 After=network.target
4
5 [Service]
6 User={{ ansible_facts['distribution'] == 'Amazon' | ternary('ec2-user', 'ubuntu') }}
7 ExecStart=/usr/bin/java -jar /home/{{ ansible_facts['distribution'] == 'Amazon' | ternary('ec2-user', 'ubuntu') }}/stockprice/
8 WorkingDirectory=/home/{{ ansible_facts['distribution'] == 'Amazon' | ternary('ec2-user', 'ubuntu') }}/
9 Restart=always
10
11 [Install]
12 WantedBy=multi-user.target
13
```

Level 4: Verification of Traffic Routing

✓ Test Nginx Proxy Forwarding

```
curl -v http://<NGINX_IP>:configured_port
```

✓ Check Java Logs

```
java -jar target/stockprice.jar
```

Next Steps

- ✓ Full automated CI/CD deployment
- ✓ Integration of Cloud Load Balancer for scalability
- ✓ Implementation of monitoring with Prometheus/Grafana